INCH-POUND

MIL-DTL-82909B(OS)
16 August 2023
SUPERSEDING
MIL-DTL-82909A(OS)
13 March 2003

# DETAIL SPECIFICATION

# BATTERIES, THERMAL, MXU-792A/A AND 792B/A

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWCIHD), and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the MXU-792A/A and MXU-792B/A Thermal Batteries, referred to herein as "battery pack".

1.2 Classification. Thermal batteries are of the following models, as specified (see 6.2).

1.2.1 Models. The models of thermal batteries are as follows:

a. Model A/A – Manufactured using chromates and European sourced materials.

b. Model B/A – Manufactured using materials sourced in the United States and uses no chromates.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-23659 | - | Initiators, Electrical, General Design Specification for |
| MIL-C-83124 | - | Cartridge Actuated Devices/Propellant Actuated Devices, General Design Specification for |
| MIL-C-83125 | - | Cartridges for Cartridge Actuated Devices/Propellant Actuated Devices, General Design Specification for |

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-202 | - | Electronic and Electrical Component Parts |

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| | | |
|---|---|---|
| MIL-STD-331 | - | Fuzes, Ignition Safety Devices and Other Related Components, Environmental and Performance Tests for |
| MIL-STD-810 | - | Environmental Engineering Considerations and Laboratory Tests |
| MIL-STD-1168 | - | Ammunition Lot Numbering and Ammunition Data Card |

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

| | | |
|---|---|---|
| MBEU145786 | - | Bracket – Battery Pack Assy |
| MBEU147699 | - | Stud Plate Assy Battery Attachment |
| MBEU147990 | - | Battery, Thermal, MXU-792A/A |
| MBEU147991 | - | Battery, Thermal |
| 838AS229 | - | MXU-792A/A Vibration/Shock Test Fixture |
| 838AS231 | - | MXU-792A/A Functional Test Setup |

(Copies of these documents are available online via Joint Engineering Data Management Information and Control System (JEDMICS) at https://webjedmics.navair.navy.mil/webjedmics/index.jsp. Refer questions, inquiries, or problems to: nani_customerservice@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

| | | |
|---|---|---|
| E1742/E1742M | - | Standard Practice for Radiographic Examination |

(Copies of this document are available online at www.astm.org.)



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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2.1 and 6.3).

3.2 Design and construction. The following shall apply to the status and configuration control of the battery pack over the entire life of the item.

3.2.1 Identical configuration. The battery pack design and construction shall be identical to that subjected to testing and service release in accordance with MIL-C-83125 and approved for service use by the Navy. The battery pack operational life shall be 225 seconds, minimum, in the operational environment of -65 to 200 °F.

3.2.2 Configuration control. Only the thermal battery procured from the vendor(s) listed on Drawing MBEU147991 is approved for the application specified. A substitute thermal battery shall not be used without prior testing and approval by the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD 20640-5035 and the Naval Air Systems Command Headquarters.

3.3 Primary components. The percussion primer and the thermal battery chemicals are primary components (see 6.4.2). Only one lot of each primary component shall be used in a battery pack lot. One primary lot may be used in more than one battery pack lot.

3.3.1 Percussion primers. Only primers manufactured within the last 24 months or recertified in accordance with the applicable primer specification within the last 12 months shall be used in the thermal battery.

3.4 Battery pack production. The battery packs shall be manufactured in accordance with Drawing MBEU147990 and the documents listed thereon. Each production battery pack shall meet the requirements of 3.3 and 3.4.1 through 3.4.8.. Failure of any battery pack to meet the requirements of 3.3 and table I shall result in rejection of that battery pack.

TABLE I. Production inspections.

| Test Sequence | Test Paragraph | Requirement Paragraph |
|---|---|---|
| Visual | 4.2.1 | 3.4.1, 3.6 |
| Radiographic Inspection | 4.2.2 | 3.4.2 |
| Leakage | 4.2.3 | 3.4.3 |
| Polarity | 4.2.4 | 3.4.4 |
| Insulation Resistance | 4.2.5 | 3.4.5 |
| Dielectric Measurement | 4.2.6 | 3.4.6 |
| Firing Pin | - | 3.4.7 |
| Actuation | 4.2.13 | 3.4.8 |



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3.4.1 Visual requirement. When the battery packs are examined in accordance with 4.2.1.1, the battery packs shall be free of the following defects: illegible, missing or inaccurate identification markings, damage, burrs, dents, sharp edges, or other defects which may prevent installation of the battery pack into its mounting bracket and gas or electrical connections. Each battery pack shall meet the requirements of 3.3, 3.6, and Drawing MBEU147990.

3.4.2 Radiographic requirement. When radiographically examined in accordance with 4.2.2, the battery pack shall show proper assembly, presence of parts, sealing and proper electrical connections, proper thread form, and presence of the battery chemicals. The radiographic inspection shall show no foreign materials present.

3.4.3 Leakage. The leakage rate of each thermal battery, prior to being installed into the battery pack manifold, shall not exceed $1.0 \times 10^{-5}$ cubic centimeter per second (cc/sec) of air when tested in accordance with 4.2.3.

3.4.4 Polarity. Each thermal battery in the battery pack shall have a polarity that matches the power and starter stack positive and negative lead configuration when tested in accordance with 4.2.4.

3.4.5 Insulation resistance. Each battery in the battery pack shall have a resistance greater than 1 megohm between the leads to case and between each set of output leads when tested in accordance with 4.2.5.

3.4.6 Dielectric measurement. Each battery in the battery pack shall not have a current drain exceeding 0.1 milliampere for a 60-second period between the leads to case and between each set of output leads when tested in accordance with 4.2.6.

3.4.7 Firing pin. Each firing pin shall have free movement (without O-ring installed) and proper alignment and protrusion in the manifold. This shall be verified during assembly of each firing pin in the manifold.

3.4.8 Actuation. The number of firing mechanisms selected from the manifold assembly production lot for the test shall be in accordance with 4.1.3.1. The retained sample shall be ignored. The mechanism shall shear between 400 to 650 pounds per square inch gauge (psig) when subjected to an onset rate of 20,000 to 40,000 psig/sec of air pressure to the inlet port when tested in accordance with 4.2.13. Failure to meet this requirement shall cause rejection of the lot. Any mechanism subjected to this test shall be handled as specified (see 6.2).

3.5 Battery pack performance. Battery packs shall meet all requirements of 3.5.1 through 3.5.8.

3.5.1 Vibration. The battery packs shall not initiate or incur any internal or external damage or degradation when vibration cycles are applied in accordance with 4.2.7.

3.5.2 Service shock. The battery pack shall not initiate or incur any internal or external damage or degradation when exposed to transient shock pulses specified in the form of shock response spectrum (SRS) from 10 to 1,000 hertz (Hz) in accordance with 4.2.8.

3.5.3 Temperature-shock/humidity/altitude (TSHA) cycling. The battery pack shall not fire or incur any internal or external degradation or damage when a TSHA cycle is applied in accordance with 4.2.9.

3.5.4 Saltfog. The battery pack shall not initiate or incur any internal or external damage nor degradation when a saltfog test is applied in accordance with 4.2.10.

3.5.5 Six-foot drop. The battery pack shall not initiate or incur any major internal or external damage or degradation as a result of the 6-foot drop test of 4.2.11.

3.5.6 Forty-foot drop. The battery pack shall not initiate or incur any internal or external damage or degradation that makes the battery pack unsafe for disposal as a result of the 40-foot drop test of 4.2.12.

3.5.7 Firing mechanism test. The battery pack shall not actuate upon an application of 400 +0, -25 psig of air pressure to the inlet port and shall actuate between 400 to 650 psig at a rate of 20,000 to 40,000 psig/sec when the firing mechanism test is applied in accordance with 4.2.13.

3.5.8 Functioning. The battery pack shall meet the requirements of 3.5.8.1 and 3.5.8.2 when fired in accordance with 4.2.14.

3.5.8.1 Rise time. Each battery in the battery pack shall be initiated and reach 22 volts within 83 milliseconds at 200 °F, within 96 milliseconds at -65 °F, and within 90 milliseconds at +70 °F.



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3.5.8.2 Current.

3.5.8.2.1 Constant. Each battery in the pack shall supply a continuous load of 0.95 ampere ±1 percent for the total battery life of 225 seconds.

3.5.8.2.2 Pulsing. Each battery in the pack shall supply 22 8-ampere ±1 percent pulses with a duration of 10 milliseconds at the times specified in table II.

TABLE II. Pulsing profile parameter.

| Pulse Number | Time (Second) | Pulse Number | Time (Second) |
|---|---|---|---|
| 1 | 0.220 | 12 | 100.15 |
| 2 | 0.320 | 13 | 100.20 |
| 3 | 0.450 | 14 | 100.21 |
| 4 | 0.650 | 15 | 160.00 |
| 5 | 0.660 | 16 | 160.15 |
| 6 | 3.110 | 17 | 160.20 |
| 7 | 60.00 | 18 | 160.21 |
| 8 | 60.15 | 19 | 224.00 |
| 9 | 60.20 | 20 | 224.15 |
| 10 | 60.21 | 21 | 224.20 |
| 11 | 100.00 | 22 | 224.21 |

3.5.8.3 Voltage. Each battery in the battery pack shall have a maximum voltage output of 60 volts and a minimum voltage output of 22 volts when tested in accordance with 4.2.14.3.

3.5.8.4 Actuation pressure. The battery pack shall initiate between 400 to 650 psig with an air application rate of 20,000 to 40,000 psig/sec.

3.5.8.5 Activation indicator. Each battery in the battery pack upon initiation shall have its activation indicator turn from white to black or pink to purple by the end of the 225 seconds.

3.6 Workmanship. The battery pack shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes, sealing, and assembly operations when tested in accordance with 4.2.1.1.

3.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).

b. Production inspection (see 4.1.2).

c. Lot acceptance inspection (see 4.1.3).

4.1.1 First article inspection. Unless otherwise specified (see 6.2), a first article sample of 39 battery packs in accordance with Drawing MBEU147990 shall be subjected to first article testing. Thirty-six of these battery packs shall be expended in the tests listed in table III and three battery packs shall be retained for investigative purposes. Any damage inflicted by the environmental treatments which would adversely affect the performance of the item in service application shall be cause for rejection of the first article sample. Any further production prior to notification by the contracting agency of first article sample acceptability is at the contractor's risk. Failure of any battery pack to comply with the requirements of section 3 shall be cause for rejection of the first article represented.



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TABLE III. First article test plan.

| Test Sequence | Test Para. | Req. Para. | Sample Group | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | I | II | III | IV | V | VI | VII | VIII | IX | X | XI |
| Visual | 4.2.1 | 3.4.1, 3.6 | 6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Radiographic | 4.2.2 | 3.4.2 | 6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Leakage | 4.2.3 | 3.4.3 | 6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Insul. Resistance | 4.2.5 | 3.4.5 | 6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Dielectric | 4.2.6 | 3.4.6 | 6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| F-18 Vibration 1/, 2/ | 4.2.7.1, 4.2.7.2 | 3.5.1 | 6 | — | — | — | — | — | — | — | — | — | — |
| T-45 Vibration 1/ | 4.2.7.3 | 3.5.1 | — | 3 | — | — | — | — | — | — | — | — | — |
| Service Shock 1/ | 4.2.8 | 3.5.2 | — | — | 3 | — | — | — | — | — | — | — | — |
| TSHA 1/ | 4.2.9 | 3.5.3 | — | — | — | 3 | — | — | — | — | — | — | — |
| Saltfog 1/ | 4.2.10 | 3.5.4 | — | — | — | — | 3 | — | — | — | — | — | — |
| Six-Foot Drop 1/ | 4.2.11 | 3.5.5 | — | — | — | — | — | 3 | — | — | — | — | — |
| Forty-Foot Drop | 4.2.12 | 3.5.6 | — | — | — | — | — | — | 3 | — | — | — | — |
| Firing Mechanism | 4.2.13 | 3.5.7 | — | — | — | — | — | — | — | 3 | — | — | — |
| Function Test: | 4.2.14 | 3.5.8 | | | | | | | | | | | |
| -65±5 °F | | | 2 | 1 | 1 | 1 | 1 | 1 | — | 1 | 3 | — | — |
| 70±5 °F | | | 2 | 1 | 1 | 1 | 1 | 1 | — | 1 | — | 3 | — |
| 200±5 °F | | | 2 | 1 | 1 | 1 | 1 | 1 | — | 1 | — | — | 3 |

NOTES:

1/ Visual, radiographic, insulation resistance, dielectric measurement inspections shall be conducted following the specific test.

2/ Footnote 1 shall be conducted after the Non-Gunfire and the Gunfire Vibration test.



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4.1.2 Production inspection. All production battery packs manufactured under the contract shall be inspected and screened for defects. Battery packs failing to meet the requirements that are listed in table I shall be rejected and removed from the lot.

4.1.3 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in table IV. Failure of any sample battery pack to comply with the requirements listed in table IV shall be cause for rejection of the lot represented.

TABLE IV. Lot acceptance inspections and tests.

| Inspection/Test | Test Paragraph | Requirement Paragraph | Quantity |
|---|---|---|---|
| Visual Inspection | 4.2.1 | 3.4.1, 3.6 | Test and retain sample |
| Radiographic Inspection | 4.2.2 | 3.4.2 | Test and retain sample |
| Leakage | 4.2.3 | 3.4.3 | Test and retain sample |
| Insulation Resistance | 4.2.5 | 3.4.5 | Test and retain sample |
| Dielectric Measurement | 4.2.6 | 3.4.6 | Test and retain sample |
| Function Test: | 4.2.14 | 3.5.8 | — |
| -65±5 °F | — | — | ⅓ Sample 1/ |
| 70±5 °F | — | — | ⅓ Sample |
| 200±5 °F | — | — | ⅓ Sample |

NOTE:

1/ Odd battery packs shall be conditioned at -65 °F.

4.1.3.1 Sample size. A random sample of battery packs from each production lot, including the samples retained for investigative purposes, shall be selected in accordance with table V for lot acceptance inspection. Test sample battery packs and samples retained for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or purchase order.

TABLE V. Lot acceptance sampling.

| Lot Size | Test Sample Size | Retained Sample Size |
|---|---|---|
| 2 – 50 | 4 | 1 |
| 51 – 90 | 6 | 1 |
| 91 – 150 | 10 | 2 |

4.2 Inspections and tests.

4.2.1 Visual inspections.

4.2.1.1 Battery pack inspection. The external condition and appearance of the battery packs shall be determined by comparison to Drawing MBEU147990. Specific inspections of the electrical connector, stainless steel battery case, aluminum manifold, and identification label shall be conducted at a minimum. Each battery pack shall meet the requirements of 3.4.1 and 3.6.



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4.2.2 Radiographic examination. Radiographic examination shall be in accordance with ASTM E1742. All battery packs and thermal batteries shall be identified with serial numbers beginning with 001 prior to examination. The battery packs and thermal batteries shall be arranged on boards or trays in consecutive order with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4- x 6-inch region, maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168 for the battery packs, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests. Any observable imperfections as outlined in 3.4.2 shall be cause for rejection of a battery pack, the thermal battery, or both. Defective battery packs or thermal batteries found during radiographic review shall be marked on the radiographic plate and removed from the production lot.

4.2.2.1 Thermal battery. Each thermal battery, prior to manifold incorporation, shall be radiographed in two directions, 90 degrees apart. Each radiograph orientation shall provide the best possible view of the internal wiring and components of the thermal battery.

4.2.2.2 Battery pack. Each battery pack, after manifold incorporation, shall be radiographed in one direction. The radiograph orientation shall be with the battery pack manifold lying flat to provide the best possible view of both internal firing mechanisms.

4.2.3 Leakage test. Leakage shall be measured after bombing in Helium gas for 60 +1.0, -0.0 minutes at a pressure of 1 +0.1, -0.0 atmospheres absolute and venting with air for 10 +2, -0.0 minutes or washed with dry nitrogen. The thermal batteries shall then be tested in a dry gas leak tester. Each thermal battery shall meet the requirements of 3.4.3.

4.2.4 Polarity test. The protective caps shall be installed on the inlet gas ports at all times during the test.

a. Each set of battery pins (S+, S-, and P+, P-) shall be tested to verify that the battery polarity conforms to the polarity indicated on the battery case. The meter used shall be capable of measurements approaching theoretical limits of sensitivity of 1,013 ohms. Prior to the start of the test, the meter used shall be reset to zero.

b. This test shall be conducted with the battery packs conditioned to 70 °F. The conditioning time shall be not less than 4 hours. Each battery pack shall be tested within 5 minutes after removal from the conditioning chamber. If there is any doubt on any polarity measurement, repeat the test after temperature conditioning at 180 °F for at least 4 hours but not more than 24 hours.

c. Pins S+ and S–

(1) On the meter, set the function switch to the “V” (millivolt) position and set the range switch to the highest range and then, upon testing, adjust to make a more precise reading.

(2) Connect the S+ and S– pins to the positive and ground connection points. The reading of the meter shall be positive or positive going.

(3) Remove the connection and reconnect to the negative and ground connection points. The reading of the meter shall be negative or negative going.

d. Pins P+ and P–

(1) On the meter, set the function switch to the “V” (millivolt position and set the range switch to the highest range and then, upon testing, adjust to make a more precise reading.

(2) Connect the P+ and P– pins to the positive and ground connection points. The reading of the meter shall be positive or positive going.

(3) Remove the connection and reconnect to the negative and ground connection points. The reading of the meter shall be negative or negative going.

e. Each battery pack shall meet the requirements of 3.4.4



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4.2.5 Insulation resistance test. The insulation resistance test outlined in MIL-STD-202, method 302 shall be used with the requirements listed below. The protective caps shall be installed on the inlet gas ports at all times.

a. Test 1.

(1) The insulation resistance shall be measured twice in test 1: (1) between the S+/S– leads in parallel to the battery case and (2) between the P+/P– leads in parallel to the battery case. Test condition "B" shall be used in accordance with method 302. The test condition shall be 500 volts (direct current) ±10 percent test potential.

(2) The insulation resistance measurement shall be made and the actual current recorded immediately after a 2-minute application of uninterrupted test voltage.

(3) However, if the instrument reading indicates that an insulation resistance meets the specified limit (1 megohm), and is steady state or increasing, the test may be terminated before the end of the 2 minutes.

b. Test 2.

(1) The insulation resistance in test 2 shall be measured in between the S+/S– (in parallel) leads and the P+/P– (in parallel) leads connected together. The test condition shall be 250 volts (direct current) ±10 percent test potential.

(2) The insulation resistance measurement shall be made immediately after a 2-minute application of uninterrupted test voltage.

(3) However, if the instrument reading indicates that an insulation resistance meets the specified limit (1 megohm), and is steady state or increasing, the test may be terminated before the end of the 2 minutes.

c. For each test conducted, the actual meter value/reading shall be recorded along with the battery pack serial number and individual thermal battery serial number.

d. Each battery pack shall meet the requirements of 3.4.5.

4.2.6 Dielectric measurement test. The dielectric withstanding voltage (DWV) test outlined in MIL-STD-202, method 301 shall be used with the requirements listed below. The protective caps shall be installed on the inlet gas ports at all times.

a. Three DWV tests shall be conducted between pin combinations: (1) between the S+/S– leads in parallel to case; (2) between the P+/P– leads in parallel to case; and (3) between the S+/S– (in parallel) and P+/P– (in parallel) leads connected together.

b. The battery insulation of each battery shall be submitted to 500 volts (alternating current) for 60 seconds. This shall be conducted by using a voltage application rate of 500 volts root-mean-square per second with the source having a nominal 60±3 Hz frequency, which shall approximate a true sine wave in form. The leakage current shall be measured with an accuracy of 5 percent.

c. For each test conducted, the actual meter value shall be recorded along with the battery pack serial number and individual thermal battery serial number.

d. Each battery pack shall meet the requirements of 3.4.5.

4.2.7 Vibration test. The battery packs shall be submitted to the non-gunfire vibration test profile: Two units at -65 °F, two units at 70 °F, and two units at 200 °F. After the non-gunfire vibration, one battery pack from each temperature group (total of three) shall be subjected to the gunfire vibration test profile at 70 °F. The test fixture shall be in accordance with Drawing 838AS229. The battery packs, with their protective plugs and caps installed, shall be vibrated along each of the three mutually perpendicular axis orientations as shown on figure 1. The battery packs shall not fire or incur internal or external damage or degradation during the vibration test. At the completion of non-gunfire vibration all battery packs shall be subjected to visual inspection and radiographic examination in accordance with 3.1 and 3.2. All battery packs shall also be subjected to the dielectric withstanding voltage and insulation resistance test in accordance with 3.2 and 3.3. The battery packs shall perform satisfactorily and meet functional testing requirements in accordance with 4.2.14.



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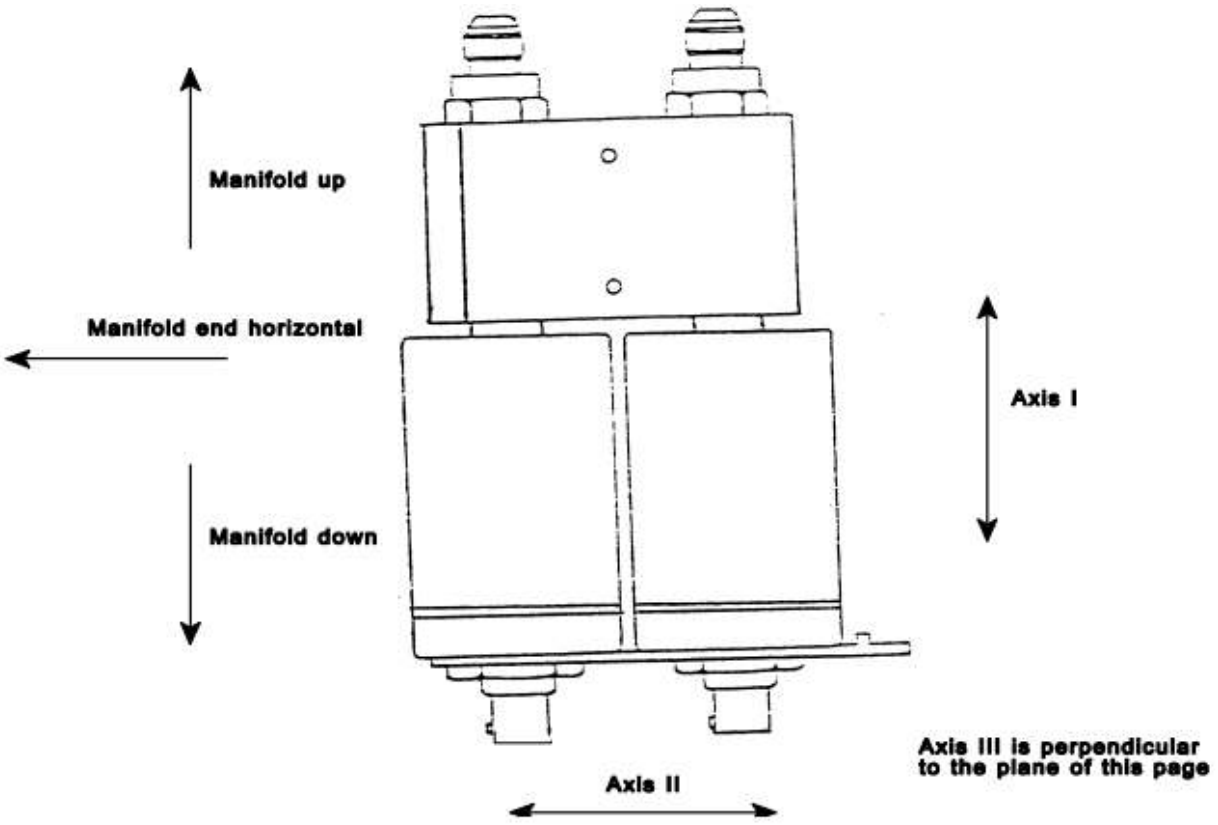



FIGURE 1. Axis orientation for the shock and vibration test and impact orientations for the six- and forty-foot drop tests.



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4.2.7.1 Non-gunfire vibration.

4.2.7.1.1 Resonance search (5 to 2,000 Hz). A resonance search shall be conducted on two of the battery packs at 70 °F in the vertical axis orientation (one at a time). The frequency sweep shall be made from 5 to 2,000 Hz at 0.024 inch double amplitude or +2 acceleration (G), whichever is less, in less than 10 minutes. The resonant points shall be noted and a transmissibility plot obtained. A resonant point shall be defined as a point that has a transmissibility (output/input) of two or greater. If more than two points are noted from the search, only the two most severe shall be used for resonance dwell. The resonant points of separate battery packs shall be considered the same if the modes are identical and the resonant frequencies fall within ±3 percent of the mean resonant frequency for each mode. If the resonant points are the same (frequencies as above and identical modes), the resonant frequency nearest the center of the 6 percent band shall be used and the battery packs vibrated simultaneously during resonance dwell. If the resonant points are not the same as defined above, the battery packs shall be vibrated separately during the dwell.

4.2.7.1.2 Resonance dwell (5 to 50 Hz). As part of the sinusoidal low frequency testing specified in 4.2.7.1.3, the battery packs shall be vibrated along the vertical axis at 70 °F for a total of 30 minutes at each of the first two resonant points. The response shall be peaked during the dwell period. The test levels for each dwell frequency shall be as shown on figure 2 at the schedule shown in table VI. Note that if there is no resonant frequency, the total cycle time shall be 90 minutes in accordance with 4.2.7.1.3. After the resonance dwell along the vertical axis, the battery packs shall be visually inspected for external damage.

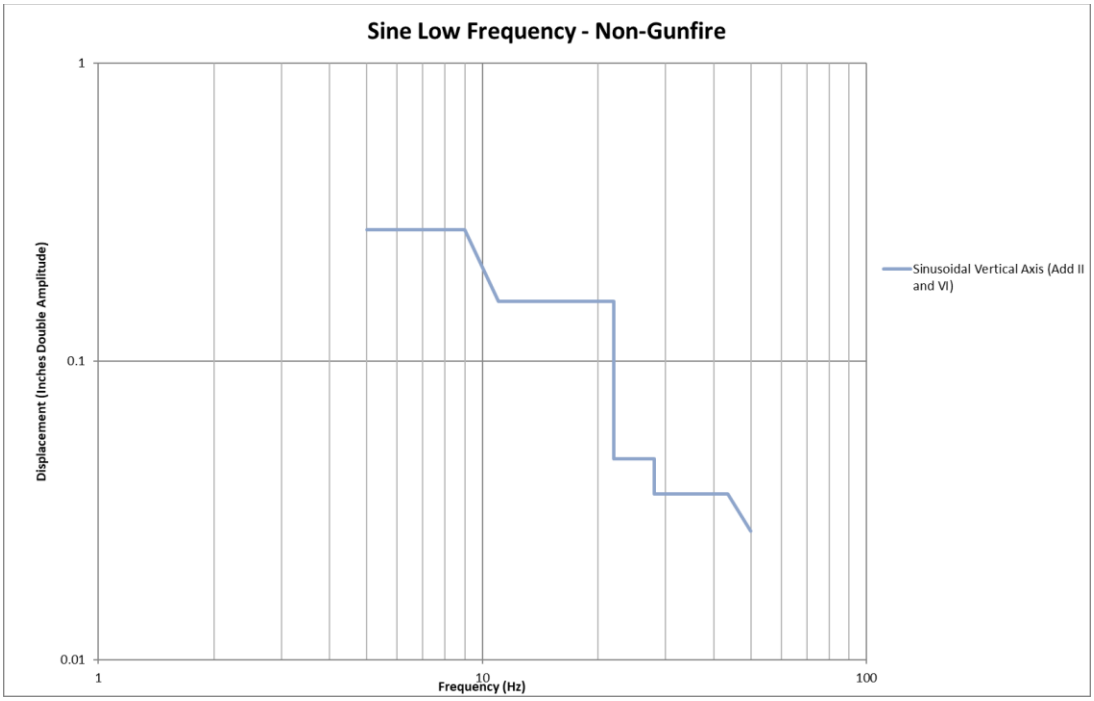



FIGURE 2. Sinusoidal low frequency vertical axis.

TABLE VI. Resonance dwell test schedule for each axis.

| Number of Resonances | | 0 | 1 | 2 |
|---|---|---|---|---|
| Total Dwell Time at Resonance Point | Endurance | 0 | 30 | 60 |
| Total Cycling Time | Endurance | 90 | 60 | 30 |



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4.2.7.1.3 Sinusoidal low frequency cycling (5 to 50 Hz). The two battery packs shall be vibrated along the vertical axis with the frequency varying at a logarithmic rate from 5 to 50 Hz and returning to 5 Hz in approximately 7.5 minutes (0.89 Octave/min). Test time shall be 90 minutes. Vibration levels shall be in accordance with figure 2 and table VII, which represent the worst case combination of F/A-18A-D and F/A-18E/F & EA-18G requirements for each axis.

TABLE VII. Sinusoidal low frequency – vertical axis.

| Vertical | | |
|---|---|---|
| **Frequency (Hz)** | **Displacement (Inches Double Amplitude)** | **Acceleration (G)** |
| 5–9 | 0.276 | — |
| 9–11 | — | ±1.1 |
| 11–22 | 0.159 | — |
| 22–28 | 0.047 | — |
| 28–43.6 | 0.036 | — |
| 43.6–50 | — | ±3.5 |
| 50 | 0.027 | — |

4.2.7.1.4 Random vibration test (50 to 2,000 Hz) – vertical axis. The two battery packs shall be vibrated along the vertical axis with the random vibration spectrum shape endurance test levels shown in figure 3 and table VIII. Frequency range for the endurance test shall be 50 to 2,000 Hz and test time shall be 120 minutes.

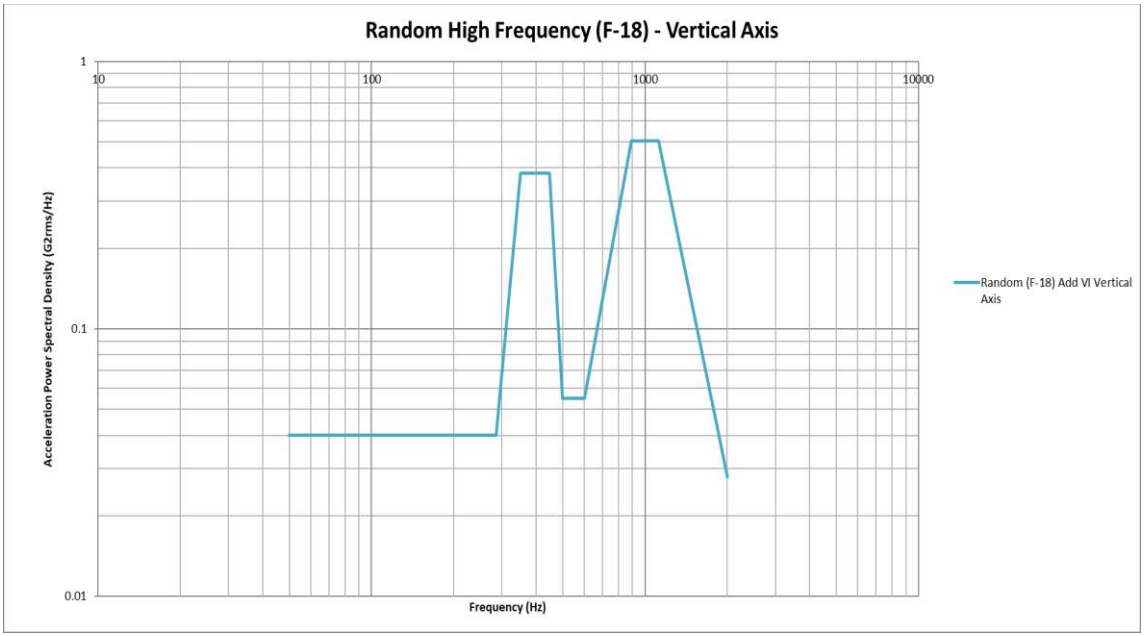



FIGURE 3. Random high frequency vertical axis.



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TABLE VIII. Random high frequency spectrum – vertical axis.

| Vertical Axis | | |
|---|---|---|
| **Frequency (Hz)** | **Acceleration ($G^2$/Hz)** | **db/Octave** |
| 50-284.9 | 0.04 | — |
| 284.9-350 | — | 33.1 |
| 350-447 | 0.383 | — |
| 447-500 | — | -52.1 |
| 500-600 | 0.055 | — |
| 600-891 | — | 16.9 |
| 891-1120 | 0.506 | — |
| 1120-2000 | — | -15.1 |
| 2000 | 0.028 | — |

4.2.7.1.5 Lateral axis. After completion of the non-gunfire vibration along the vertical axis the procedures above, including resonance search and dwell, shall be repeated along the lateral axis orientation, except that the sinusoidal vibration levels shall be vibrated as shown in figure 4 and table IX and the random vibration spectrum endurance test levels shall be vibrated as shown in figure 5 and table X.

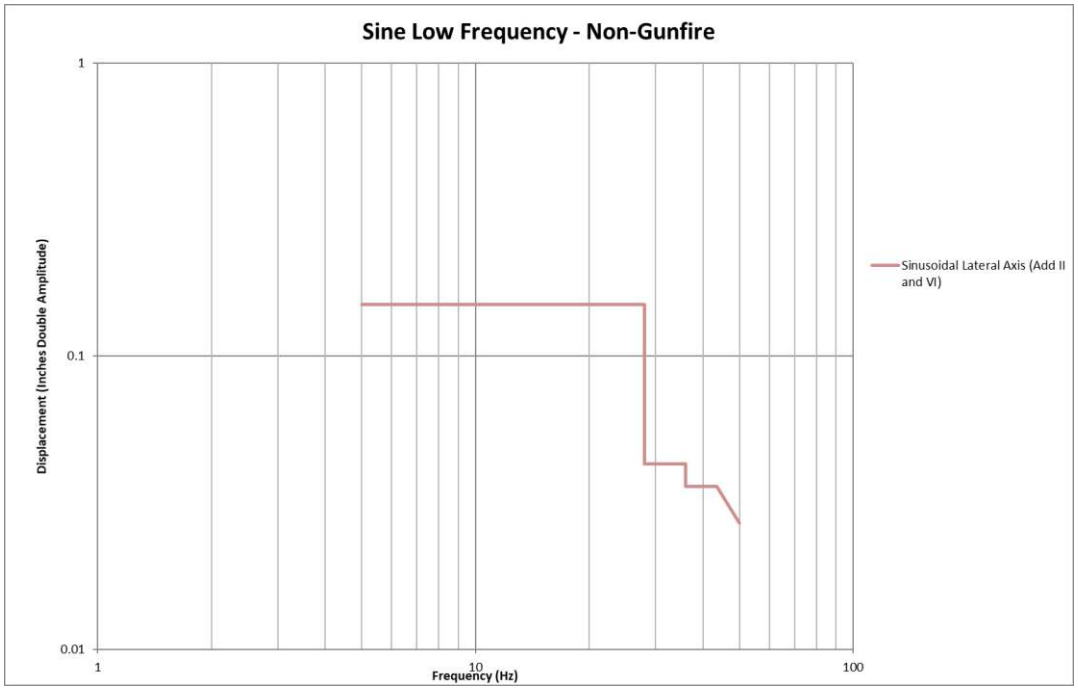



FIGURE 4. Sinusoidal low frequency lateral axis.



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TABLE IX. Sinusoidal low frequency – lateral axis.

| Lateral Axis | | |
|---|---|---|
| **Frequency (Hz)** | **Displacement (Inches Double Amplitude)** | **Acceleration (G)** |
| 5–28 | 0.149 | — |
| 28–36 | 0.043 | — |
| 36–43.6 | 0.036 | — |
| 43.6–50 | — | ±3.5 |
| 50 | 0.027 | — |

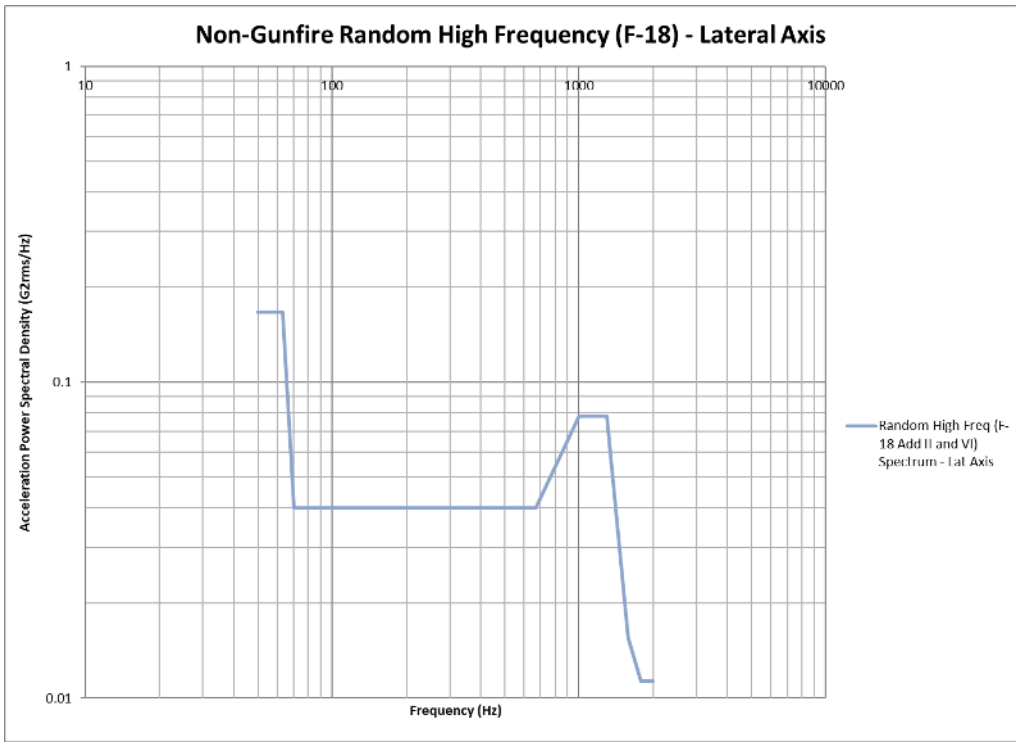



FIGURE 5. Random high frequency lateral axis.



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TABLE X. Random high frequency – lateral axis.

| Lateral Axis | | |
|---|---|---|
| **Frequency (Hz)** | **Acceleration ($G^2$/Hz)** | **db/Octave** |
| 50–63 | 0.167 | — |
| 63–70.1 | — | -40.3 |
| 70.1–670 | 0.04 | — |
| 670–1,000 | — | 5 |
| 1,000–1,300 | 0.078 | — |
| 1,300–1,580 | — | -24.9 |
| 1,580–1,789 | — | -7.7 |
| 1,789–2,000 | 0.011 | — |

4.2.7.1.6 Longitudinal axis. After completion of the non-gunfire vibration along the lateral axis the procedures, including resonance search and dwell, shall be repeated along the longitudinal orientation, except that the sinusoidal vibration levels shall be vibrated as shown in figure 6 and table XI and the random vibration spectrum endurance test levels shall be vibrated as shown in figure 7 and table XII.

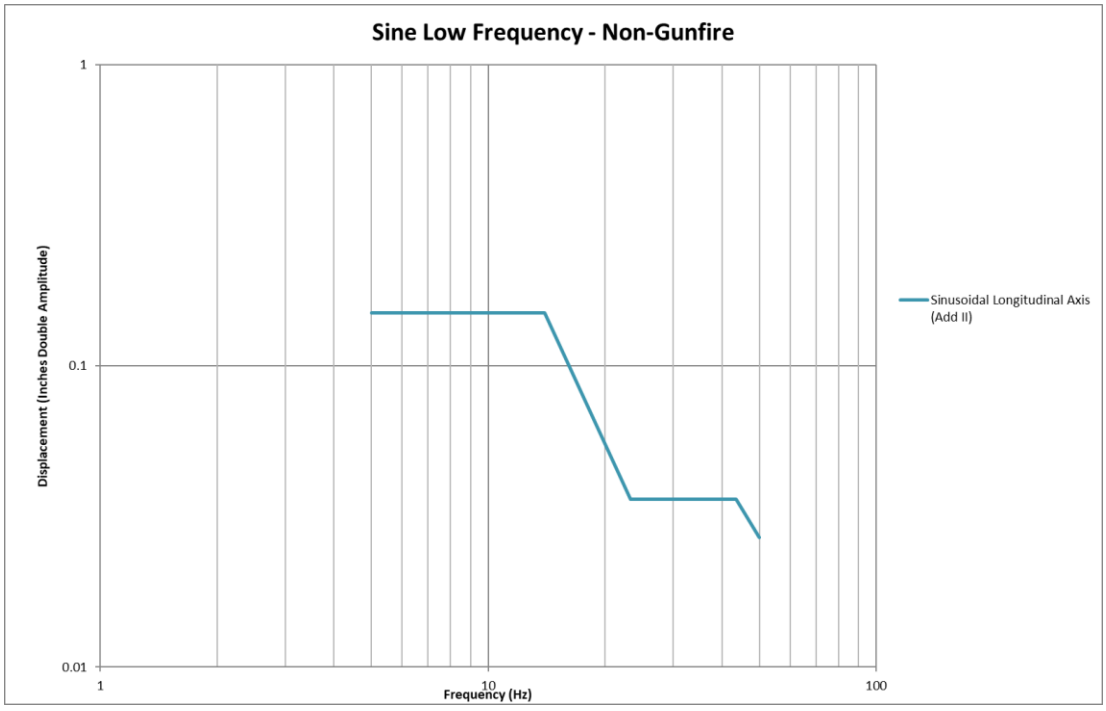



FIGURE 6. Sinusoidal low frequency longitudinal axis.



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TABLE XI. Sinusoidal low frequency – longitudinal axis.

| Longitudinal Axis | | |
|---|---|---|
| **Frequency (Hz)** | **Displacement (Inches Double Amplitude)** | **Acceleration (G)** |
| 5–14 | 0.15 | — |
| 14–23 | — | ±1 |
| 23.3–43.6 | 0.036 | — |
| 43.6–50 | — | ±3.5 |
| 50 | 0.027 | — |

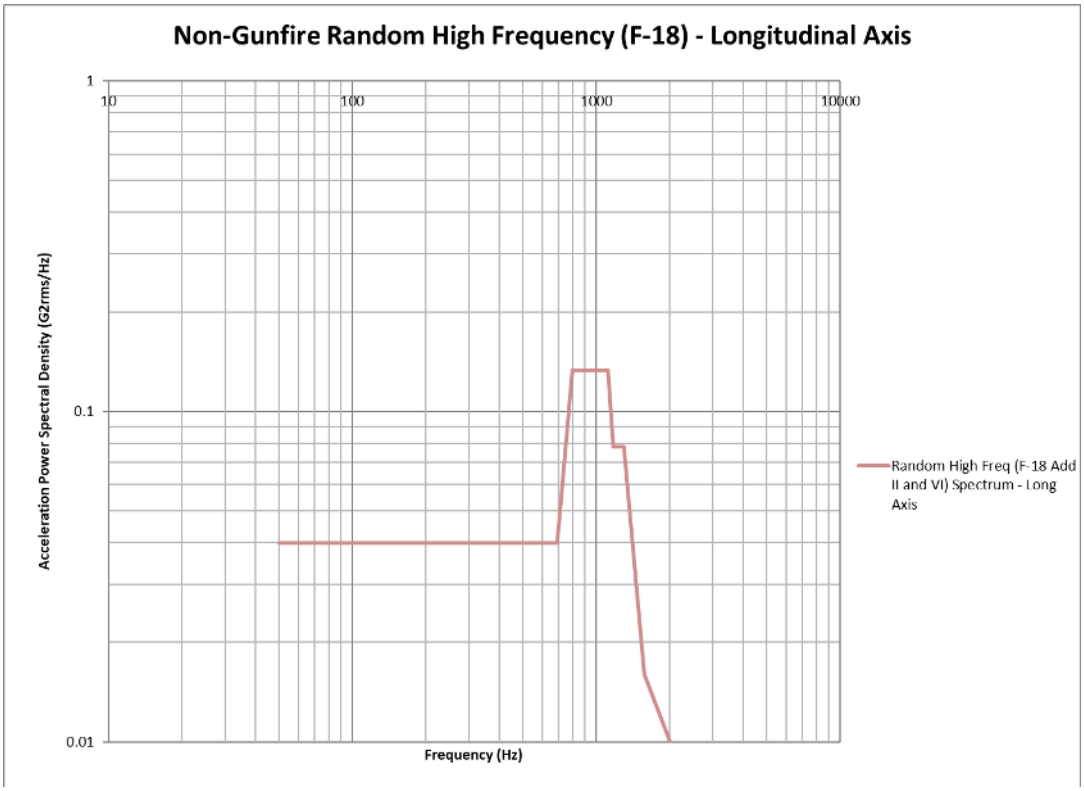



FIGURE 7. Random high frequency longitudinal axis.



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TABLE XI. Random high frequency – longitudinal axis.

| Longitudinal axis | | |
|---|---|---|
| **Frequency (Hz)** | **Acceleration ($G^2$/Hz)** | **db/Octave** |
| 50–691.5 | 0.04 | — |
| 691.5–800 | — | 24.8 |
| 800–1,120 | 0.133 | — |
| 1,120–1,174.7 | — | -33.5 |
| 1,174.7–1,300 | 0.078 | — |
| 1,300–1,574 | — | -24.9 |
| 1,574 | 0.016 | — |
| 1,547–2,000 | — | -5.9 |
| 2,000 | 0.01 | — |

4.2.7.1.7 -65 °F non-gunfire vibration. After completion of the non-gunfire vibration at +70 °F, the procedures outlined above shall be repeated with two new battery packs and the temperature shall be -65 °F in lieu of +70 °F.

4.2.7.1.8 +200 °F non-gunfire vibration. At the completion of the non-gunfire vibration at -65 °F, the procedures outlined above shall be repeated with two new battery packs and the temperature shall be +200 °F in lieu of +70 °F.

4.2.7.2 F-18 gunfire vibration. The gunfire vibration test is performed in the frequency range of 50 to 2,000 Hz. Testing shall consist of logarithmic sweep and narrow band sweep tests in accordance with the following paragraphs. The gunfire tests shall be performed on each battery pack (one from each non-gunfire vibration temperature group) along all three orthogonal axes at +70 °F.

4.2.7.2.1 Logarithmic sweep. The three battery packs shall be vibrated along the vertical axis, with the frequency varying at a logarithmic rate from 50 to 2,000 Hz and back to 50 Hz in approximately 15 minutes (0.71 octaves per minute). Double amplitude or vibratory acceleration levels shall be in accordance with figure 8 and table XIII. Test time shall be 60 minutes.



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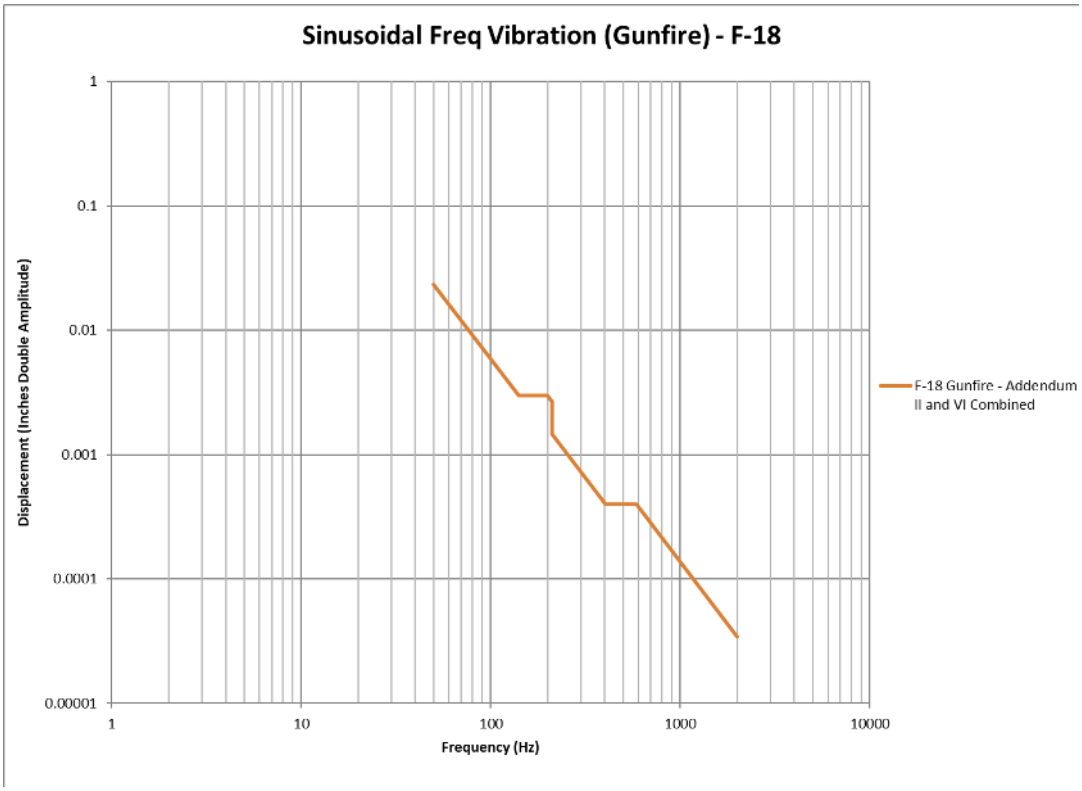



FIGURE 8. Vibration (gunfire) – sinusoidal spectrum.

TABLE XIII. Vibration (gunfire) – sinusoidal test schedule.

| F-18 Gunfire – Addendum VI | | |
|---|---|---|
| **Frequency (Hz)** | **Displacement (Inches Double Amplitude)** | **Acceleration (G)** |
| 50 | 0.0235 | — |
| 50–140 | — | ±3 |
| 140–200 | 0.003 | — |
| 200–210 | — | ±6 |
| 210 | 0.0027-0015 | — |
| 210–402 | — | ±3.3 |
| 402–590 | 0.0004 | — |
| 590–2,000 | — | ±7.1 |
| 2,000 | 0.0000343 | — |



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4.2.7.2.2 Narrow band sweep. The three battery packs shall be vibrated along the same orthogonal axis, at the fixed narrow band sweep frequencies. The fixed narrow band sweeps shall be performed by sweeping the frequencies about ±5 percent of the specified center frequency; see table XIV for frequency and sweep times. The sweep rate shall be approximately the same as the logarithmic sweep above (0.71 octaves per minute). Test time shall be 90 minutes.

TABLE XIV. Vibration (gunfire) – narrow band sweep schedule.

| Center Frequency | ±5% | Duration (min) per axis | Level (g) |
|---|---|---|---|
| 67 | (63.7–70.4) | 7.5 | 3.3 |
| 100 | (95–105) | 7.5 | 3.3 |
| 135 | (128–142) | 7.5 | 3.3 |
| 200 | (190–210) | 15 | 3.3 |
| 267 | (254–280) | 7.5 | 3.3 |
| 300 | (285–315) | 7.5 | 3.3 |
| 335 | (318–352) | 7.5 | 3.3 |
| 400 | (380–420) | 15 | 3.3 |
| 467 | (444–490) | 7.5 | 4.5 |
| 500 | (475–525) | 7.5 | 5.1 |
| | Total | 90 | |



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4.2.7.2.3 Lateral axis. After completion of the gunfire vibration along the vertical axis, the procedures shall be repeated along the lateral axis orientation.

4.2.7.2.4 Longitudinal axis. After completion of the gunfire vibration along the lateral axis, the procedures shall be repeated along the longitudinal axis orientation.

4.2.7.3 T-45 Vibration.

4.2.7.3.1 Random vibration – vertical. One battery pack shall be vibrated along the vertical axis at +70 °F with the combined random vibration spectrum shape of the endurance test and minimum structural integrity levels in accordance with figure 9 and table XV. Frequency range for the endurance test shall be 5 to 2,000 Hz and test time shall be 60 minutes.

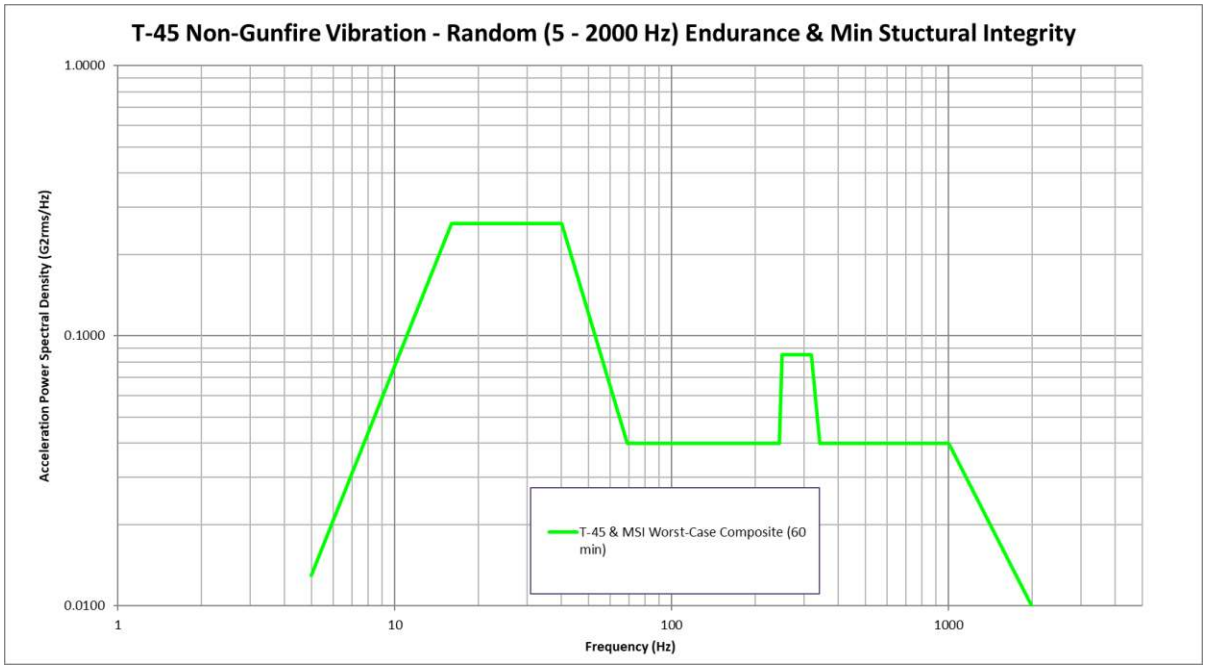



FIGURE 9. T-45 non-gunfire random vibration.

TABLE XV. T-45 random vibration test schedule.

| Longitudinal Axis | | |
|---|---|---|
| **Frequency (Hz)** | **Acceleration ($G^2$/Hz)** | **db/Octave** |
| 5 | 0.013 | — |
| 5–16 | — | 7.75 |
| 16–40 | 0.26 | — |
| 40–69 | — | -10.3 |
| 69–245 | 0.04 | — |
| 245–250 | — | 112 |
| 250–320 | 0.085 | — |
| 320–342 | — | -34.1 |



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TABLE XV. T-45 random vibration test schedule – Continued.

| Longitudinal Axis | | |
|---|---|---|
| **Frequency (Hz)** | **Acceleration ($G^2$/Hz)** | **db/Octave** |
| 342–1,000 | 0.04 | — |
| 1,000–2,000 | — | -6.02 |
| 2,000 | 0.01 | — |

4.2.7.3.2 Lateral axis. After completion of the non-gunfire vibration along the vertical axis, the above procedures shall be repeated along the lateral axis orientation.

4.2.7.3.3 Longitudinal axis. After completion of the gunfire vibration along the lateral axis, the above procedures shall be repeated along the longitudinal axis orientation.

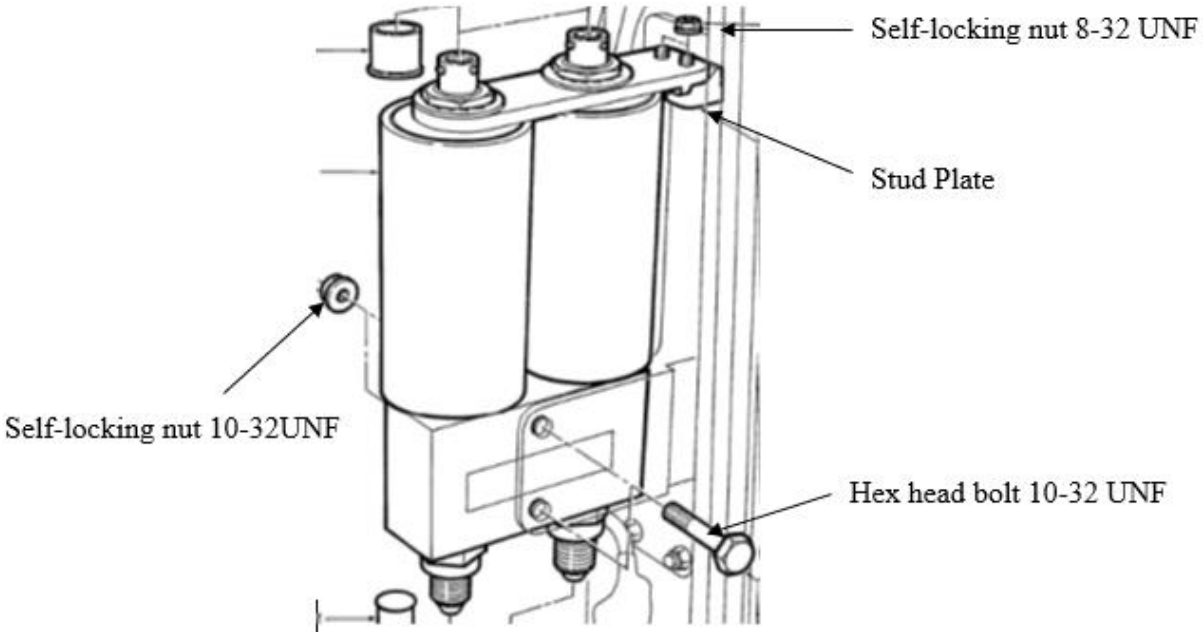
4.2.7.3.4 -65 °F Non-gunfire vibration. After completion of the non-gunfire vibration at +70 °F, the procedures outlined above shall be repeated with a new battery pack and the temperature shall be -65 °F in lieu of +70 °F.

4.2.7.3.5 +200 °F Non-gunfire vibration. At the completion of the non-gunfire vibration at -65 °F, the procedures outlined above shall be repeated with a new battery pack and the temperature shall be +200 °F in lieu of +70 °F.

4.2.8 Service shock test. A different SRS profile shall be tested in each orthogonal axis. Eighteen shocks at ambient temperature shall be conducted, six applied to each profile. For each profile, three shocks shall be applied followed by three shocks where the sign of the synthesized time history is reversed or the required peak acceleration is applied in the opposite direction. Figure 10 shows the mounting attachments of the battery to the seat mounting bracket (MBEU145786). The shock test input and the shock response spectrum for the lateral, longitudinal, and vertical axes (orientation shown on figure 1) shall be conducted as shown on figures 11 through 14. The battery pack shall be installed into the seat mounting bracket at four mounting points. Two hex head bolts, 10-32 UNF (NAS6303U19) and self-locking nuts, 10-32 UNF (NAS1291C3M) shall be installed to connect the seat mounting bracket at the thermal battery manifold and torqued to 35 inch-pounds. Two self-locking nuts, 8-32 UNF (NAS1291C08M) shall also be torqued to 25 inch-pounds to the stud plate (MBEU147699) to interface the seat mounting bracket to the battery bracket.



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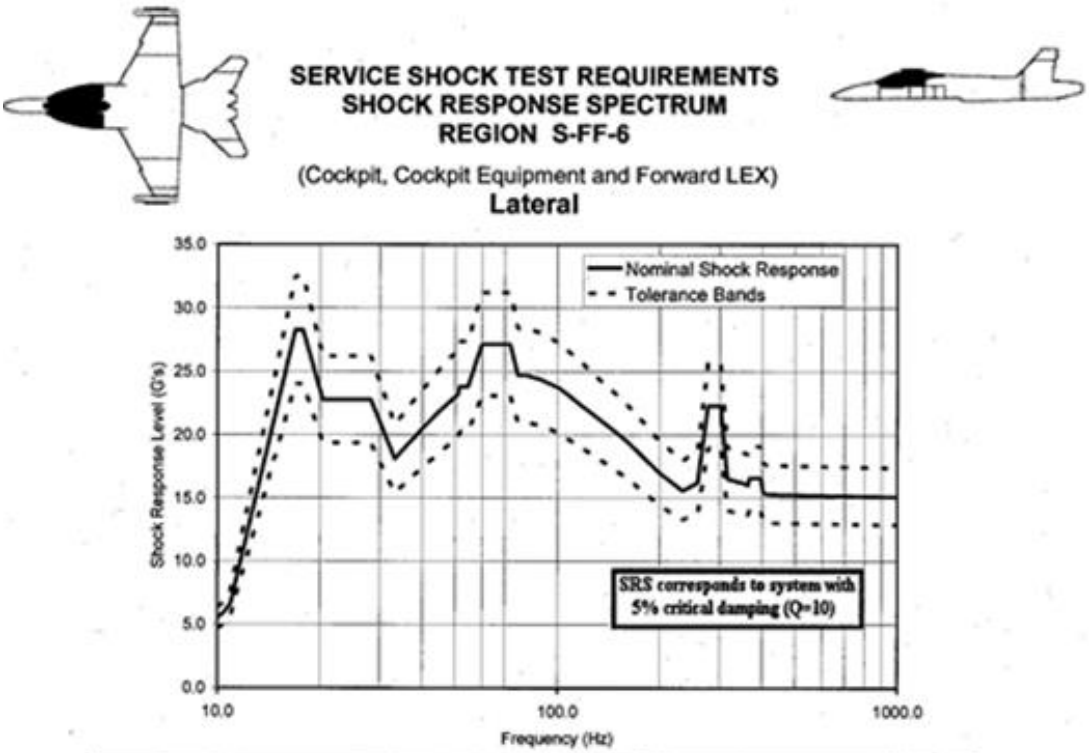

NOTE:

The seat mounting bracket shall then be attached to the shock/vibration test plate using screw head bolts (MS16997-34) torqued to 35 inch-pounds. After each shock, an evaluation shall be performed to determine if the required test shock response spectrum, test waveform peak G and test waveform effective time duration ($T_E$) is within tolerance. Figure 14 defines the test effective shock duration. The battery packs shall not initiate or incur any internal or external damage or degradation during service shock. The battery packs' performance shall not be affected by the service shock test.

FIGURE 10. Thermal battery attachment to the seat mounting bracket.



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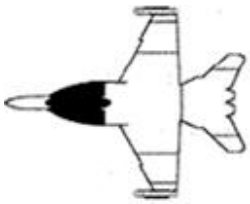

**Service Shock Region S-FF-6**
**Shock Response Levels (Lateral)**

| Frequency (Hz) | Nominal (G's) | Upper Tolerance (G's) | Lower Tolerance (G's) | Frequency (Hz) | Nominal (G's) | Upper Tolerance (G's) | Lower Tolerance (G's) |
|---|---|---|---|---|---|---|---|
| 10.0 | 5.5 | 6.3 | 4.7 | 81.0 | 24.7 | 28.4 | 21.0 |
| 10.5 | 5.9 | 6.8 | 5.1 | 89.0 | 24.4 | 28.1 | 20.7 |
| 10.9 | 6.6 | 7.6 | 5.6 | 101.0 | 23.7 | 27.3 | 20.1 |
| 16.9 | 28.3 | 32.5 | 24.0 | 157.0 | 19.7 | 22.7 | 16.7 |
| 17.8 | 28.3 | 32.5 | 24.0 | 201.0 | 16.9 | 19.4 | 14.4 |
| 20.3 | 22.8 | 26.2 | 19.4 | 233.0 | 15.6 | 17.9 | 13.3 |
| 28.2 | 22.8 | 26.2 | 19.4 | 258.5 | 16.2 | 18.6 | 13.8 |
| 33.2 | 18.1 | 20.8 | 15.4 | 277.1 | 22.3 | 25.6 | 19.0 |
| 37.0 | 19.5 | 22.4 | 16.6 | 301.7 | 22.3 | 25.6 | 19.0 |
| 45.0 | 21.9 | 25.2 | 18.6 | 316.2 | 16.5 | 19.0 | 14.0 |
| 49.0 | 22.8 | 26.2 | 19.4 | 357.0 | 16.1 | 18.5 | 13.7 |
| 51.0 | 23.2 | 26.6 | 19.7 | 362.7 | 16.0 | 18.4 | 13.6 |
| 51.4 | 23.8 | 27.4 | 20.2 | 367.4 | 16.6 | 19.1 | 14.1 |
| 54.5 | 23.8 | 27.4 | 20.2 | 396.5 | 16.6 | 19.1 | 14.1 |
| 59.9 | 27.1 | 31.2 | 23.1 | 402.0 | 15.4 | 17.8 | 13.1 |
| 72.6 | 27.1 | 31.2 | 23.1 | 413.0 | 15.3 | 17.6 | 13.0 |
| 76.4 | 24.7 | 28.4 | 21.0 | 1000.0 | 15.1 | 17.4 | 12.8 |

FIGURE 11. Lateral SRS profile region S-FF-6 – peak acceleration = 15.1 g minimum, effective test duration ($T_E$) = 0.118 (sec) maximum.



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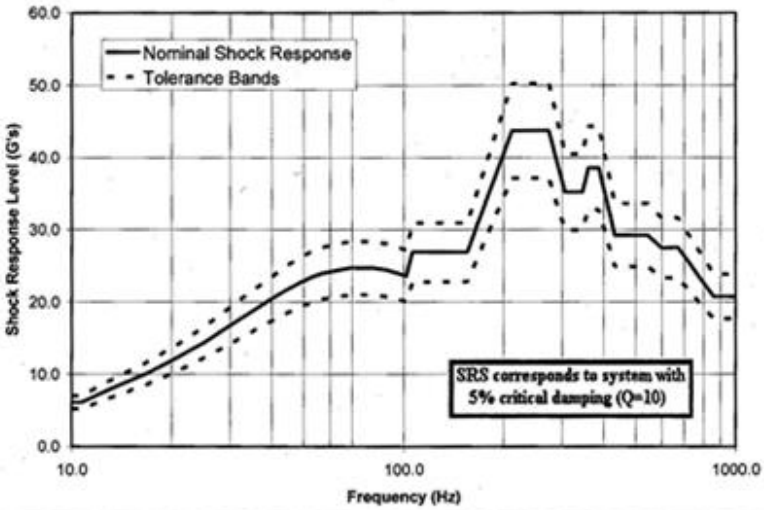


**SERVICE SHOCK TEST REQUIREMENTS**
**SHOCK RESPONSE SPECTRUM**
**REGION S-FF-6**

(Cockpit, Cockpit Equipment and Forward LEX)

**Longitudinal**



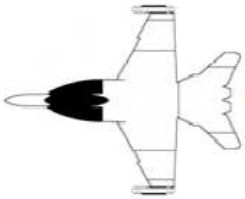



| Service Shock Region S-FF-6 Shock Response Levels (Longitudinal) | | | | | | | |
|---|---|---|---|---|---|---|---|
| Frequency (Hz) | Nominal (G's) | Upper Tolerance (G's) | Lower Tolerance (G's) | Frequency (Hz) | Nominal (G's) | Upper Tolerance (G's) | Lower Tolerance (G's) |
| 10.0 | 6.1 | 7.0 | 5.2 | 101.8 | 23.6 | 27.2 | 20.1 |
| 10.7 | 6.1 | 7.0 | 5.2 | 106.2 | 26.9 | 30.9 | 22.9 |
| 13.0 | 7.9 | 9.1 | 6.7 | 155.5 | 26.9 | 30.9 | 22.9 |
| 17.0 | 10.2 | 11.7 | 8.7 | 211.8 | 43.7 | 50.3 | 37.2 |
| 21.0 | 12.4 | 14.3 | 10.5 | 273.4 | 43.7 | 50.3 | 37.2 |
| 25.0 | 14.4 | 16.6 | 12.2 | 305.8 | 35.2 | 40.5 | 29.9 |
| 29.0 | 16.3 | 18.7 | 13.9 | 345.1 | 35.2 | 40.5 | 29.9 |
| 33.0 | 18.0 | 20.7 | 15.3 | 359.3 | 38.5 | 44.3 | 32.8 |
| 37.0 | 19.5 | 22.4 | 16.6 | 386.0 | 38.5 | 44.3 | 32.8 |
| 45.0 | 21.9 | 25.2 | 18.6 | 431.7 | 29.3 | 33.7 | 24.9 |
| 49.0 | 22.8 | 26.2 | 19.4 | 542.5 | 29.3 | 33.7 | 24.9 |
| 53.0 | 23.5 | 27.0 | 20.0 | 598.7 | 27.5 | 31.6 | 23.3 |
| 57.0 | 24.0 | 27.6 | 20.4 | 669.7 | 27.5 | 31.7 | 23.4 |
| 69.0 | 24.7 | 28.4 | 21.0 | 856.8 | 20.8 | 23.9 | 17.6 |
| 81.0 | 24.7 | 28.4 | 21.0 | 1000.0 | 20.8 | 23.9 | 17.6 |
| 89.0 | 24.4 | 28.1 | 20.7 | | | | |

FIGURE 12. Longitudinal SRS profile region S-FF-6 – 20.8 g minimum, effective test duration ($T_E$) = 0.029 (sec) maximum.



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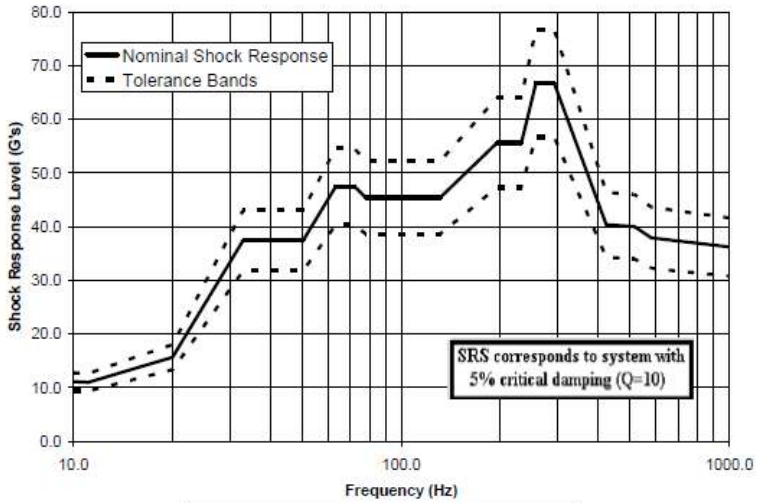


**SERVICE SHOCK TEST REQUIREMENTS**
**SHOCK RESPONSE SPECTRUM**
**REGION S-FF-6**

(Cockpit, Cockpit Equipment and Forward LEX)

**Vertical**






| Service Shock Region S-FF-6 Shock Response Levels (Vertical) | | | |
|---|---|---|---|
| Frequency (Hz) | Nominal (G's) | Upper Tolerance (G's) | Lower Tolerance (G's) |
| 10.0 | 11.1 | 12.8 | 9.5 |
| 11.1 | 11.0 | 12.7 | 9.4 |
| 20.1 | 15.7 | 18.0 | 13.3 |
| 33.0 | 37.4 | 43.0 | 31.8 |
| 50.3 | 37.4 | 43.0 | 31.8 |
| 63.2 | 47.5 | 54.6 | 40.4 |
| 72.0 | 47.5 | 54.6 | 40.4 |
| 78.0 | 45.4 | 52.2 | 38.6 |
| 131.5 | 45.4 | 52.2 | 38.6 |
| 196.2 | 55.6 | 64.0 | 47.3 |
| 232.7 | 55.6 | 64.0 | 47.3 |
| 257.9 | 66.7 | 76.7 | 56.7 |
| 293.0 | 66.7 | 76.7 | 56.7 |
| 422.2 | 40.3 | 46.4 | 34.3 |
| 514.1 | 40.0 | 46.0 | 34.0 |
| 580.0 | 37.9 | 43.6 | 32.2 |
| 1000.0 | 36.2 | 41.6 | 30.8 |

FIGURE 13. Vertical SRS profile S-FF-6 – peak acceleration = 36.2 g minimum, effective test duration ($T_E$) = 0.061 (sec) maximum.



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**MAXIMUM ALLOWABLE TEST EFFECTIVE SHOCK DURATION, $T_E$**

(Calculated on typical synthesized input pulse waveform)

Tolerance On Specified $T_E$ Value: $\pm$ 10%

$T_E$ is defined to be the minimum length of time that contains all time history magnitudes exceeding in absolute value one-third of the shock peak magnitude absolute value, $(\frac{A_{pk}}{3}, \frac{-A_{pk}}{3})$, associated with the shock input waveform.

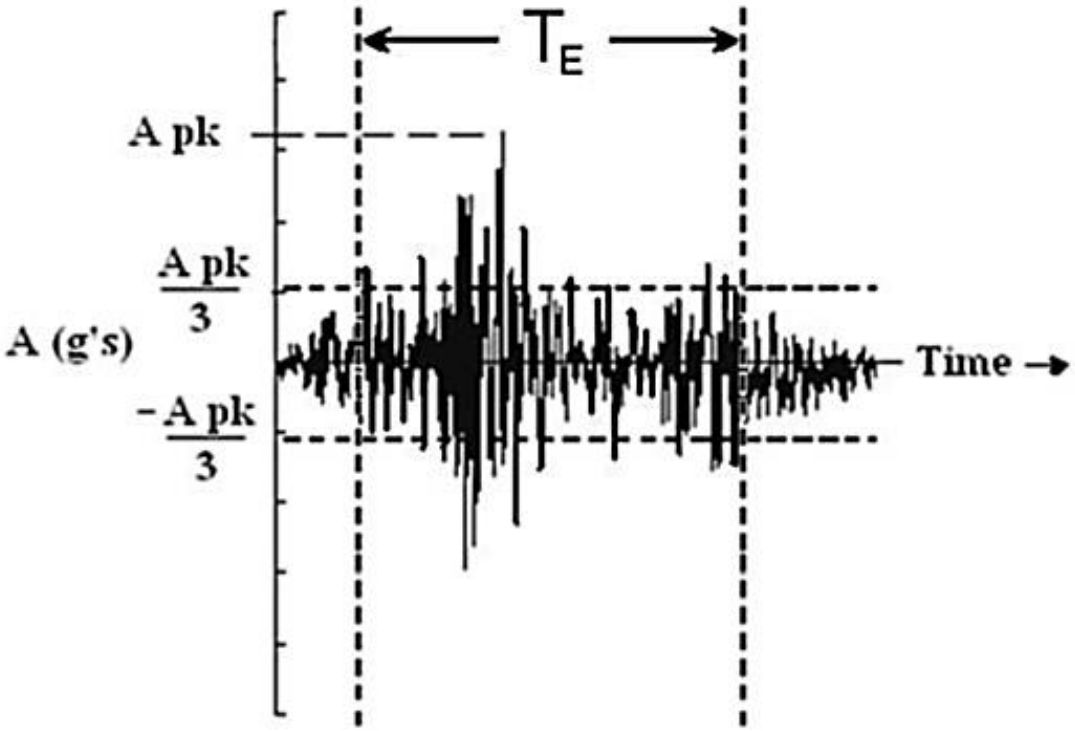



FIGURE 14. Test effective shock duration.

4.2.9 TSHA cycling test. The battery packs with protective plugs installed in the inlet and electrical ports shall be submitted to the TSHA cycling test outlined in MIL-DTL-23659 with the following requirements:

a. The battery packs shall be supported in such a way that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. No battery pack shall be touched by another battery pack during the cycling schedule shown in table XVI.

b. Failure of any battery pack to comply with the requirements of 3.5.3 shall be cause for rejection of the lot or first article.



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TABLE XVI. TSHA cycling schedule.

| | | |
|---|---|---|
| Monday | 0800 | Place battery packs in a chamber at 70 °F at 50% RH [1/]. |
| | 1200 | Raise chamber temperature to 160 °F and the RH to 95%. The chamber temperature shall reach 160 °F at 95% RH no later than 1300. |
| | 1600 | Remove battery packs from chamber and immediately place in chamber maintained at -65 °F at a pressure altitude of 70,000 feet - (0.65 psi [2/]). |
| Tuesday | 0800 | Remove battery packs from chamber and immediately place in chamber maintained at 70 °F at 50% RH. |
| | 1200 | Remove battery packs from chamber and immediately place in chamber maintained at -65 °F at a pressure altitude of 70,000 feet - (0.65 psi). |
| | 1600 | Remove battery pack from chamber and immediately place in a chamber maintained at 160 °F at 95% RH. |
| Wednesday | 0800 | Reduce chamber temperature to 70 °F and RH to 50%. The chamber temperature shall reach 70 °F at 50% RH no later than 0900. |
| | 1200 | Raise chamber temperature to 160 °F and RH to 95%. The chamber temperature shall reach 160 °F at 95% RH no later than 1300. |
| | 1600 | Remove battery packs from chamber and immediately place in chamber maintained at -65 °F at a pressure altitude of 70,000 feet - (0.65 psi). |
| Thursday | 0800 | Remove battery packs from chamber and immediately place in chamber maintained at 70 °F at 50% RH. |
| | 1200 | Remove battery packs from chamber and immediately place in chamber maintained at -65 °F at a pressure altitude of 70,000 feet - (0.65 psi). |
| | 1600 | Remove battery pack from chamber and immediately place in a chamber maintained at 160 °F at 95% RH. |



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TABLE XVI. TSHA cycling schedule – Continued.

| | | |
|---|---|---|
| | 0800 | Reduce chamber temperature to 70 °F and RH to 50%. The chamber temperature shall reach 70 °F at 50% RH no later than 0900. |
| | 1200 | Raise chamber temperature to 160 °F and RH to 95%. The chamber temperature shall reach 160 °F at 95% RH no later than 1300. |
| Friday | 1600 | Remove battery pack from chamber and immediately place in a chamber maintained at -65 °F at standard ambient pressure. |

FOOTNOTES:

1/ Relative humidity.

2/ Pound-force per square inch.

NOTE: This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekend the soak time shall be from 1200 on Friday until 0800 Monday at a temperature of 160 °F at 95 percent relative humidity.

4.2.10 Salt fog test. The battery packs, with protective plugs installed in the inlet and electrical ports, shall be submitted to the salt fog test outlined in MIL-STD-810, method 509 with the following requirements:

a. Waxed string shall be used to support the battery packs in the chamber. The battery pack shall be oriented in the manifold up position as shown on figure 1. No change in orientation is required during the test. The battery packs shall be exposed to a 5 percent salt solution for 168 hours. Condensed fog shall be collected at two points every 12 hours and the specific gravity, concentration, and PH of the solution measured.

b. After completing the 168 hours of testing, the salt deposits (if any) shall be removed, if necessary, for aiding in a visual inspection. To remove the salt deposits, a gentle washing technique shall be used with water not warmer than 100 °F.

c. After completing the salt fog test, the battery packs shall be stored for 48 hours at 70 °F. At the end of the 48 hours, a complete visual inspection shall be conducted. Any damage or degradation shall be noted.

d. Failure of any battery pack to comply with the requirements of 3.5.4 shall be cause for rejection of the lot or first article.

4.2.11 Six-foot drop test. The battery packs, with protective plugs removed from the inlet ports and electrical connectors, shall be submitted to the drop test outlined in MIL-C-83125 with the following requirements:

a. The battery packs shall be dropped from a height of 6 feet onto a 2-inch-thick steel plate supported by reinforced concrete. One new battery pack shall be used for each drop orientations. All impact orientations are given on figure 1.

b. After each battery pack is dropped, a visual inspection shall occur to document any minor or major external damage that occurred and to verify non-initiation of either battery.

c. If it is determined that the damage sustained by any one battery pack cause it to be unsafe for handling, all testing shall be stopped.

d. Failure of any battery pack to comply with the requirements of 3.5.5 shall be cause for rejection of the lot or first article.

4.2.12 Forty-foot drop test. The battery packs, with protective plugs removed from the inlet ports and electrical connectors and without shipping containers, shall be submitted to the drop test outlined in MIL-STD-331, test 103 with the following requirements:

a. The battery packs shall be dropped from a height of 40 feet onto a 2-inch-thick steel plate supported by reinforced concrete. One new battery pack shall be used for each drop orientations. All impact orientations are given on figure 1.



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b. After each battery pack is dropped, a visual inspection shall occur to document any external damage that occurred and to verify non-initiation of either battery.

c. If it is determined that the damage sustained by any one battery pack cause it to be unsafe for handling (ruptured case), all testing shall be stopped.

d. At the conclusion of the forty-foot drop test, each battery pack shall again be visually inspected for external damage. After being inspected, each battery pack shall then be disposed of in a safe and appropriate manner.

e. Failure of any battery pack to comply with the requirements of 3.5.6 shall be cause for rejection of the lot or first article.

4.2.13 Firing mechanism test. The battery packs shall be submitted to the firing mechanism test outlined in MIL-C-83124 with the following requirements:

a. The battery packs shall be divided into three groups. Each battery pack group shall be assigned a conditioning temperature of one of the following:
   (1) -65 °F.
   (2) 70 °F.
   (3) 200 °F.

The conditioning time intervals of the group assigned -65 or 200 °F shall be at least 4 hours, but not more than 24 hours at the group's assigned conditioning temperature. The conditioning time interval of the group assigned 70 °F shall be conditioned for at least 4 hours at 70 °F. Each group shall be conditioned to their assigned conditioning temperature prior to functional testing. Each test shall start within 3 minutes from conditioning chamber removal.

b. Non-Actuation: A pressure of 400 +0, -25 psig air pressure shall be gradually applied through a 0.013-inch orifice to one of the ports of each battery pack (check valve removed) for 1 minute. The other inlet port shall be fitted with a pressure transducer to monitor leakage past the check valve.

c. Actuation: A pressure of 650 psig air pressure shall be applied at a rate of approximately 20,000 to 40,000 psig/sec to one port of each battery pack. The other inlet port shall be fitted with a pressure transducer to monitor leakage past the check valve.

d. Failure of any battery pack to comply with the requirements of 3.5.7 shall be cause for rejection of the lot or first article.

4.2.14 Functional test. Sample battery packs for the functional test shall be conditioned for not less than 4 hours and no more than 24 hours at the specified temperatures. The battery packs shall be removed from the conditioning chamber and test fired within 5 minutes. If any battery pack is not fired within 5 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The battery packs shall be initiated using an activation pressure of 900 +25, -0 psig of air. The battery pack shall be fired in the battery pack test firing setup shown on Drawing 838AS231. The battery packs shall meet the requirements specified in 3.5.8.

4.2.14.1 Rise time. The rise time shall be measured from the primer ignition (600 psig) to when the battery voltage reaches 22 volts when measured before the test firing circuit diodes. Each battery pack shall meet the requirements of 3.5.8.1.

4.2.14.2 Current. A constant and pulsing currents shall be measured over the first 225 seconds. The battery pack shall meet the requirements of 3.5.8.2.

4.2.14.3 Voltage. The four voltages shall be measured over the first 225 seconds before the diodes indicated on Drawing 838AS231, test firing circuit. Each battery pack shall meet the requirements of 3.5.8.3.

4.2.14.4 Test failure. If a test failure is attributable to an assignable cause, excluding the battery packs, the original test results shall be discarded and the test reconducted.



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Classification (see 1.2).

c. When first article inspection is required (see 3.1 and 4.1.1).

d. When mechanisms subjected to testing are required to be discarded or certified acceptable prior to being reworked (see 3.4.8).

e. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the U.S. Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the U.S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Flight critical part. A part, the single failure of which, during any operating condition, could cause loss of aircraft or one of its major components, loss of control, unintentional release of, or inability to release, or which may cause significant personnel injury, including during flight, escape, survival, or rescue. The battery pack is considered a flight critical part and the technical data package is to be identified and procured as such.

6.4.2 Primary components. Primary components are all components in which a functional failure would result in a misfire or malfunction of the battery pack.



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6.5 Part Numbers. MBEU145786, MBEU147699, MBEU147990, MBEU147991

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6.6 Subject term (key word) listing.

Chemicals
Current
Polarity
Primer
Voltage

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 6135-2023-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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